Data Science Assignment #1 **Ananya Subramaniam and Sree Lasya Gogineni**

IDEA A: Evolution of Popular Music

1. Data Collection
    a. This collection uses observation and measurement. We will use observation because we are watching and tracking behaviors over time. This includes **date, artists, and genres**. Additionally, we will measure songs' **duration, BPM, and chart position,** and compare them with other songs to see what has changed. We are asking the research question: "**To what extent have the characteristics of popular songs changed over time?"** The data collection need is driven by trying to understand trends in popular music over the years and potentially compare them across ages. We want to show similar features that have stayed the same, while also emphasizing the defining features of culturally relevant songs.
    b. Data Management Plan
        i. Creation of logical collections
            1. For logical collections, we will primarily organize them by year. We will display them in 5-year increments to group the songs together. Genre will also serve as a subgrouping to show further distinctions. Our main columns will be year, artist, genre, duration, BPM, and chart positions. By analyzing these attributes, we can see if characteristics can affect rankings and popularity.
        ii. Physical data handling
            1. We can use Billboard Top 100 rankings, which go back to 1963. A GitHub repository called Historic Billboard Hot 100 Data also provides exportable JSON files that can be used to extract data. These JSON files have been included every week since 1958. The GitHub repository provides the date, song, artist, the song's current chart position, the song's previous-week chart position, the song's peak chart position, and the number of weeks the song has been on the chart. The first Kaggle dataset includes the same data, which will be used to verify consistency. We have to use another Kaggle dataset, the Spotify Tracks Dataset, to get the genre, BPM, and duration. This allows us to track and relate the metadata.
        iii. Interoperability support
            1. We will store the collected Billboard music data in platform-independent, widely supported formats such as CSV and JSON, allowing use across Windows, macOS, and Linux systems. Because the GitHub repository and Kaggle are both open source, anyone can access them. We will maintain a consistent data structure for attributes such as **date, song title, artist, genre,**

duration, BPM, and chart position. We will also maintain a data dictionary and consistent metadata conventions so the fields can be correctly interpreted.

iv. Security support
    1. Regarding security, neither the repository nor Kaggle is serving any personal information. Confidentiality is not a problem, as both are open source. Additionally, GitHub has its own security logs to track data vulnerabilities. The public, including us, do not have access to the actual data collected; it is just openly available.

v. Data ownership
    1. As mentioned in the security section, access to the repository and all data is public, but changes aren't available unless you own the repository or dataset. There are no limitations on access and use. You can verify data quality by cross-checking the repository and the Kaggle dataset. Also, for the Billboard Top 100 Tracks Kaggle dataset, there are auto updates done every Wednesday at 2 pm which eliminates human error. Replication can be used to verify data points and get more info by using both Kaggle and GitHub.

vi. Metadata collection, management, and access
    1. Since the official Billboard website has data from 1963 to now and the GitHub repository has data from 1958, we will use data from both starting in 1963 so we can cross-verify accuracy. This will help us avoid incomplete and inconsistent data. Also, by including genre, duration, and BPM from the Spotify Tracks Dataset, we are looking more closely at the song itself. Identifying these threads will help us spot potential underlying trends.

vii. Persistence
    1. We can add the link to the public GitHub repository. We can also add the links to Billboard's Hot 100 and Spotify Tracks Dataset Kaggle datasets. Since we are not generating new data, we are not adding any unique data. However, we can add our findings and conclusions in a GitHub repository so people who want to learn more can access them.

viii. Knowledge and information discovery
    1. This will be available in a public GitHub repository. If someone finds the dataset and needs a particular file, the Historic Billboard Hot 100 Data repository has folders organized by date. Each week has a separate JSON file, and a README explains each dataset's contents and purpose. The Kaggle datasets, The Billboard Hot 100

and Spotify Tracks Dataset, are available in Python and can be downloaded as a zip file.

ix. Data distribution and publication

1. The data we use is the Billboard Hot 100 Rankings, which is updated every week. We also use the GitHub repository, which is updated weekly. For the metadata, we are using the Spotify Tracks dataset, and we will also add the link so that the data is accessible. This ensures the data stays up to date. However, our findings will be limited to 1963 to the date we present, since we won't continuously update the project.

2. Survey of data storage/formats
    a. We can use the JSON files from the public GitHub repository. JSON is useful for preserving this hierarchical relationship between a chart date and the multiple songs belonging to that chart. We can also download the Billboard Top 100 Rankings as a zip file, unzip the file, and then open it as a CSV. CSV is helpful because each chart observation can be a row and its attributes can be columns, making the data easy to process with tools such as Python and R. For our dataset, the JSON records could be transformed into a CSV structure in which each row represents one song's appearance on a particular Billboard chart date, making comparisons across decades easier.
3. Survey of metadata conventions, standards
    a. Our metadata is primarily based on background information about the songs themselves. This includes genre, duration, and BPM. This metadata is coming from Spotify Tracks Kaggle Dataset. Dublin Core is the standard used for metadata. The following metadata follows the Dublin Core standard and covers the data in our GitHub dataset, Kaggle Dataset for Billboard Hot 100, and Kaggle Dataset for Spotify Tracks :
        i. Title: Evolution of Popular Music Using Billboard Hot 100 Data
        Creator: mhollingshead
        Subject: Popular Music, Billboard Hot 100
        Format: JSON
        Date: XX
        Source: Billboard dataset
        ii. Kaggle Dataset 1
            1. Title: Billboard Hot 100 & More
            Creator: Godefroy Lambert
            Subject: Billboard Historical Charts
            Format: CSV
            Date: XX
            Source: Billboard dataset

iii. Kaggle Dataset 2

1. Title: Spotify Tracks Dataset
   Creator: Yash - creator of the dataset
   Subject: Information about Spotify tracks across 125 different music genres
   Format: CSV
   Date: XX
   Source: Spotify dataset

4. Provenance - Just Ananya for 6000 level
   a. The provenance for our data follows this chain: Original source → Raw data → Cleaning → Transformation → Final dataset → Analysis. Our original source is the GitHub repository and the Kaggle CSVs. We are accessing it throughout the fall 2026 semester - September through December. When we clean it, we want to choose songs that accurately represent our 5-year segments. We transform the JSON into a CSV and clean the missing values as well. Lastly, we add the additional attributes we obtained from the metadata.

IDEA B: Internships at RPI

1. Data Collection
   a. For this collection, we would use measurement and observation. We are measuring RPI students and the relationship between their internships and the school's resources. The question we are asking is "**How do students' academic experiences and university resources relate to the quality and outcomes of their internship experiences?"** The variables we are looking at include the student, their specific internship experience, major relevance, RPI resources, and their desired outcome. Most of these variables will be collected using multiple-choice questions, numerical ranges, and rating scales so that responses can be compared quantitatively. Some of the questions will also allow for optional open-ended responses if an individual wants to add more details to their answer. Data collection is driven by a research question and a research idea to understand which academic and university-related resources students use and how they relate to their work experiences and career prospects.
   b. Data Management Plan
      i. Creation of logical collections
         1. For logical collections, we will primarily organize them by class year like freshman, sophomore, junior and senior. We will also separate into subgroups based on majors and industries that people are working in. This way it is easier to understand and see patterns even within different majors and industries as well as different class years. By doing this, it gives people the ability to quickly sort

and find the data they are looking for based on the parameters they are looking for.

ii. Physical data handling
    1. Data is coming from surveys. Via Google Forms, we will convert the survey data into a Google Sheet which can become a CSV. Our data will be largely formatted using ranges, numeric conventions, yes/no, and multiple-choice questions. The naming conventions will follow ASCII and Database rules.

iii. Interoperability support
    1. The data that is being collected will be stored in CSV, so that it is easier to parse. This data will also be added to a GitHub repository so that the data will be easy to access. We will maintain a consistent data structure for attributes listed in the below table. We will also maintain a data dictionary and consistent metadata conventions so the fields can be correctly interpreted.

Category	**Data Point**
Student	Major
	Academic Level
	GPA Range
Internship	Industry
	Internship role / Category
	Internship Duration
	Hours Per Week
	Work Arrangement
Compensation	Was the internship paid?
	Hourly Pay
	How related was the internship to your major?
	Was this the type of career you expected to pursue?
Academic Preparation	Did coursework prepare you?
	How much of what you learned in class was applicable?

	Most useful skills
	Were there skills you needed but had not learned in class?
RPI Resources	How did you find the internship?
	Did RPI resources contribute to obtaining it?
	Which RPI resources did you use?
Experiences / Outcomes	Overall internship satisfaction
	Did the internship meet expectations?
	Did you receive a return offer?
	Would you work for the company again?
	Do you want to continue in this type of work?
	Did the internship change your career goals?

iv. Security support
    1. To secure our data, we will save our results in a CSV and upload it to a GitHub repository. If this data changes, it will be because more people filled out our survey, and we will need to update the static CSV from the Google Sheets document. Only we will have access to make changes. We will also preserve student anonymity because we won't collect personal information such as names and RINs.

v. Data ownership
    1. Ananya and Sree Lasya will collect the data and ensure data quality. This means that when we make the questionnaire to send out we will make sure there are no leading questions, have spaces to add additional information, and correct ranges and answer choices. This data will be uploaded to a repository in which only Ananya and Sree Lasya will have access to change/modify the data so not everyone can have access to change anything. We will create documentation explaining each field we consider. We will also add the list of survey questions and steps to replicate the study so other people can conduct it in different colleges. In the future, this data can be used to draw comparisons between different colleges on the same objective measures.

vi. Metadata collection, management, and access

1. The metadata is that all data is collected from RPI students, including all undergraduate students in their freshman, sophomore, junior, or senior year. We will also include questions with an optional space to explain further. We will be reading through each response and making sure that the responses are related and are complete and consistent with the answer choice that they selected. This metadata will be stored in an additional column that keeps track of the optional, extra information that someone writes.

vii. Persistence
1. The data will be available in our repository; however, the Google Form will not be accessible. Though we will make the questions known, keeping the form static allows us to maintain the integrity of our results. After the class/semester/graduation, it might be good to repeat this study because many of these factors change as time continues. Other factors to take into consideration are time periods, available resources, national unemployment rates, and many more will come up as we continue this project.

viii. Knowledge and information discovery
1. People can find our data in the repository. The fields will be specified in the CSV. Additionally, we will have documentation specifying each field. These will have their definition and relevance to the overall project. In the data, the variables will be viewed as column names. In collections and files, the variables will be specified once, almost like a key, to be referenced throughout the report.

ix. Data distribution and publication
1. There will likely be no major data updates as there are no plans to continue this project after this semester. However, if there are changes, we would have to implement a notification system to make sure people interested are aware of the new information. To track use and significance, we can look at interactions with the Github repository. If there are people interested in this study, they can use the resources in the GitHub repository to replicate this project, and can contact us if there are any additional questions.

2. Survey of data storage/formats
a. The primary data format we will be using is CSV. Survey responses will initially be collected using Google Forms and stored in Google Sheets, where each row will represent an individual survey response and each column will represent their answers to the questions specified above. The Google Sheet can then be exported as a CSV file, which is appropriate because most of our collected data is

structured and consists of categorical values, numerical ranges, Yes/No responses, and rating scales. A way to expand this is by making some of the responses into binary fields using 0 for no and 1 for yes. We will also maintain a separate data dictionary describing each variable, its data type, possible values, and meaning so that the CSV data can be correctly interpreted by other users.

3. Survey of metadata conventions, standards
    a. Our metadata is primarily based on the population we are collecting the data from. The population is RPI students in their freshman, sophomore, junior, or senior year. We are also collecting optional responses where participants can explain their answers in more detail. We will vet these responses to ensure the information is consistent and complete. We can apply the Dublin Core standard to account for metadata information regarding title, date of collection, and subject (major). To take into account metadata for each individual question, we can apply the Data Documentation Initiative (DDI) standard. This way, we can take into account the wording of each question, the different categories of responses, and the valid value ranges. DDI will also help with optional open-ended responses by giving us a structured way to document them and ensure consistency and completeness.
4. Provenance - Ananya
    a. The data we are collecting comes directly from RPI students as they answer questions about their specific internship experience. The original data will come directly from our Google Forms survey, and we will record the survey collection period, survey version, and the questions that were presented to participants. We will preserve the original Google Forms responses and Google Sheet as raw data, and create a separate CSV file for cleaning and analysis. We will conduct this project throughout the Fall 2026 semester, but we will stop the survey a few weeks before our final report to ensure the data remains consistent. Documentation will make the study easier to reproduce if we conduct a similar study again.